5.3 Technical Requirements

- Python as the primary programming language.
- Appropriate Python libraries for data analysis and machine learning.
- A suitable development environment.
- A machine-learning model appropriate to the selected prediction problem.
- A suitable dataset for training and testing the model.
- Sufficient computing resources to develop and demonstrate the prototype.

5.4 User Requirements

The proposed system is intended to provide useful information to healthcare staff and managers. Users should therefore be able to understand the information presented by the system without needing to understand the underlying programming or machine-learning algorithms.

The system should present predictions and results clearly so that they can be interpreted as decision-support information.

6. Constraints

6.1 Data Availability

The quality and availability of healthcare data may limit the performance of the AI model. A dataset that is too small, incomplete or not representative of the intended environment may affect the reliability of the predictions.

6.2 Data Privacy

Healthcare information can contain sensitive patient information. For the purposes of the student project, the group must avoid exposing personally identifiable patient information. The use of publicly available or appropriately anonymised data is therefore important.

6.3 Project Time

The solution must be developed within the timeframe provided for the Business Analysis 3.2 project. This limits the complexity and scope of the system that can realistically be developed and tested.

6.4 Technical Skills

The project is being developed by students, which may limit the complexity of the machine-learning techniques, software features and healthcare functionality that can be implemented within the available time.

6.5 Computing Resources

The performance and complexity of the AI model may be limited by the available computers and software environments used by the group.

6.6 Scope of the Prototype

PREDICT-CARE is a student project and therefore represents a prototype rather than a fully deployed healthcare information system. The project will demonstrate the concept and technical feasibility of the solution rather than provide a production-ready clinical system.

7. Risks

Risk	Possible Impact	Mitigation
Poor-quality or incomplete data	The model may produce unreliable predictions.	Clean the dataset and identify/remove or appropriately handle missing and inconsistent data.
Insufficient dataset size	The model may not learn patterns effectively.	Select a suitable dataset and evaluate whether it contains sufficient relevant observations.
Model overfitting	The model may perform well on training data but poorly on unseen data.	Separate training and testing data and use appropriate model-evaluation techniques.
Inaccurate predictions	Incorrect predictions could reduce the usefulness of the system.	Evaluate the model using appropriate performance measures and compare suitable algorithms where possible.

Risk	Possible Impact	Mitigation
Data privacy concerns	Sensitive patient information could be exposed.	Use anonymised or publicly available data and avoid including personally identifiable information.
Technical difficulties	Problems with Python, libraries or the development environment could delay development.	Test the development environment early and maintain backups of project files.
Limited project time	The group may not complete all planned features.	Prioritise the core AI functionality and avoid unnecessary features.
Misinterpretation of AI results	Users may treat predictions as medical decisions.	Clearly communicate that PREDICT-CARE is a decision-support tool and does not replace professional medical judgement.

8. Tools and Techniques

PREDICT-CARE will use Python as the primary programming language.

8.1 Python

Python will be used to develop the data-processing, machine-learning and prediction components of PREDICT-CARE.

8.2 Data Cleaning and Preprocessing

Data preprocessing techniques will be used to prepare the dataset before it is supplied to the machine-learning model. This may include:

- Handling missing values.
- Removing duplicate records.
- Correcting inconsistent data.
- Converting data into suitable formats.

- Selecting relevant variables.
- Preparing categorical and numerical data for modelling.

8.3 Exploratory Data Analysis

Exploratory Data Analysis (EDA) will be used to understand the dataset and identify patterns, relationships, trends and unusual values before developing the machine-learning model.

8.4 Machine Learning

A suitable machine-learning algorithm will be selected based on the type of prediction problem being addressed. The model will be trained using the prepared dataset and tested using unseen data.

The selection of the final algorithm will be based on its suitability for the selected problem and its performance during evaluation.

8.5 Model Evaluation

The machine-learning model will be evaluated using appropriate performance measures. The evaluation method will depend on whether the final prediction problem is treated as a classification or regression problem.

The purpose of model evaluation will be to determine whether the model can produce reliable predictions and whether improvements or alternative algorithms are required.

8.6 Data Visualisation

Graphs and other visualisations may be used to present patterns and results from the dataset. Visualisation will help users understand trends in patient demand, waiting times and other relevant variables.

8.7 GitHub

GitHub will be used to store and manage the group's project work.

Overall Business Objective

Overall, PREDICT-CARE aims to demonstrate how Artificial Intelligence can be applied within the healthcare industry to transform available data into useful predictions and insights. The solution focuses on supporting patient management, improving operational planning and assisting healthcare staff with data-driven decision-making. It is therefore intended to provide a practical example of how AI can be used to address an industry-specific problem while remaining achievable

within the scope of the Business Analysis 3.2 project.